

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

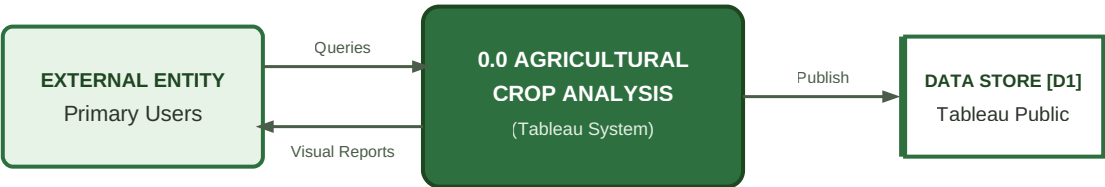
Date: 13 July 2026
Team ID: SB-2026-AGRI
Max Marks: 4 Marks

Project Title: India's Agricultural Crop Production Analysis using Tableau
Domain: Agricultural Data Analytics & Visualization (SmartBridge Capstone)

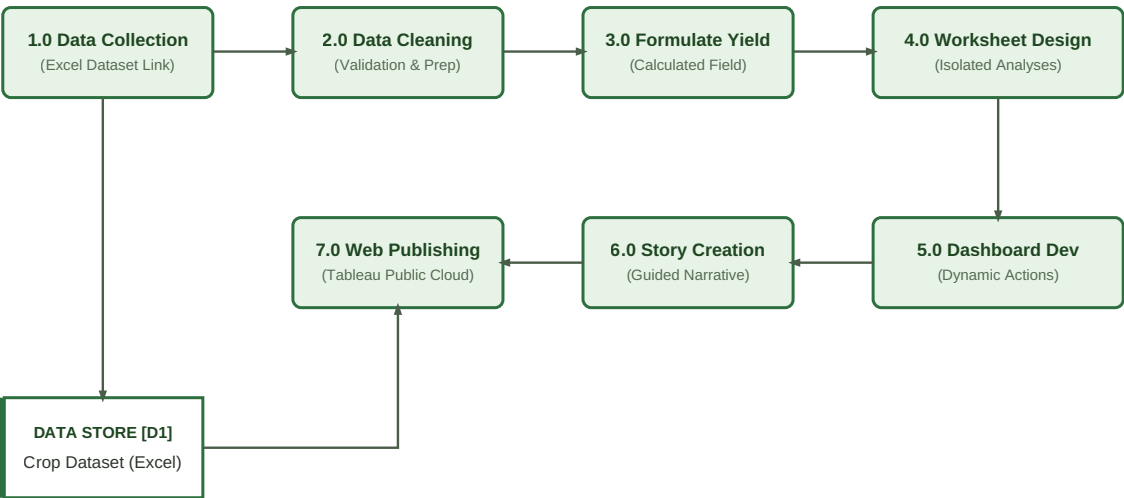
1. Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

LEVEL-0 CONTEXT DIAGRAM



LEVEL-1 DATA FLOW DIAGRAM (TABLEAU PIPELINE)



2. Data Flow Diagram Explanation

The logical architecture of the **India's Agricultural Crop Production Analysis** system flows sequentially through systematic stages of data processing and presentation inside Tableau.

Initially, the system reads raw agriculture production records from the Microsoft Excel workbook data store. The data cleaning module verifies region dimensions and strips null variables. A custom calculated field is immediately configured within Tableau to formulate the crop **Yield** (defined as Production divided by Area), producing a fair baseline for cross-state comparison.

These cleaned dimensions are plotted across specialized worksheets (seasonal trends, crop comparison tables, geographical maps). The sheets are arranged onto a main dashboard layout using active action filters. The resulting sequential narrative is compiled into a Tableau Story and deployed straight onto the Tableau Public server. From this cloud environment, farmers, policy makers, and analysts can seamlessly extract high-value insight.

3. User Stories

The following tabular breakdown traces specific system capabilities requested by distinct actors participating in the agricultural analytics domain, strictly baseline aligned to the final deployment on **13 July 2026**.

USER TYPE	FUNCTIONAL EPIC	ID	USER STORY / TASK DESCRIPTION	ACCEPTANCE CRITERIA	PRIORITY	RELEASE
Analyst	Data Connection	USN-1	As an analyst, I want to bind the crop database into Tableau so that analytical schemas are instantly established.	Successful connection with clean data typing across all 10 schema fields.	High	13 July 2026
Developer	Derived Metrics	USN-2	As a developer, I want to build a calculated field for Yield to compare crop productivity fairly.	Yield displays accurately as: [Production] / [Area] without zero-division errors.	High	13 July 2026
Officer	Regional Mapping	USN-3	As an officer, I want a regional map visualization to identify highly productive zones at a glance.	Interactive filled map clearly scales colors based on total regional production.	High	13 July 2026
Researcher	Seasonal Trends	USN-4	As a researcher, I want to filter metrics by crop year and season to evaluate periodic patterns.	Line charts immediately update based on active multi-select dashboard filters.	Medium	13 July 2026
Policy Maker	Unified Dashboard	USN-5	As a policy maker, I want a single dashboard with big KPI cards to evaluate critical national crop yield summaries.	The dashboard clusters production, area, and yield dynamically across selection panels.	High	13 July 2026
Public / Student	Web Presentation	USN-6	As a student, I want to view a published storyboard online so I can follow the analysis guide.	Tableau Public link successfully loads a responsive, multi-page web story layout.	High	13 July 2026